

CO3 - AT1 - Analytical Problem solving

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QUESTIONS

Q1
CO3 - AT1 - Analytical Problem Solving
20 marks**Q2**
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CO3 - AT1 - Analytical Problem Solving
20 marks**Q1** CO3 - AT1 - Analytical Problem Solving

20 marks

1. Given an $N \times N$ matrix. Write a C Program to determine whether it is:

- a) Symmetric
- b) Upper triangular
- c) Lower triangular
- d) Diagonal matrix
- e) Identity matrix
- f) Display all applicable classifications.

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,a[20][20],i,j,s=1,u=1,l=1,d=1,id=1;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10        for(j=0;j<n;j++) {
```

Input

```
3
1 0 0
0 1 0
0 0 1
```

Output

Solving
20 marks**Q4**
CO3 - AT1 - Analytical Problem Solving
20 marks**Q5**
CO3 - AT1 - Analytical Problem Solving
20 marks

5/5 answered

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,a[20][20],i,j,s=1,u=1,l=1,d=1,id=1;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10        for(j=0;j<n;j++) {
11            if(a[i][j]!=a[j][i])
12                s=0;
13            if(i>j && a[i][j]!=0)
14                u=0;
15            if(i<j && a[i][j]!=0)
16                l=0;
17            if(i==j && a[i][j]!=0)
18                d=0;
19            if(i==j && a[i][j]!=1)
20                id=0;
21            if(i!=j && a[i][j]!=0)
22                id=0;
23        }
24     if(s)
25         printf("Symmetric\n");
26     if(u)
27         printf("Upper Triangular\n");
28     if(l)
29         printf("Lower Triangular\n");
30     if(d)
31         printf("Diagonal matrix\n");
32     if(id)
33         printf("Identify matrix\n");
34     return 0;
35 }
36 }
```

Input

```
3
1 0 0
0 1 0
0 0 1
```

Output

```
Symmetric
Upper Triangular
Lower Triangular
Diagonal matrix
Identify matrix
```

Run

Save

Solving
20 marks

Q4 ✓
CD3 - AT1 - Analytical Problem
Solving
20 marks

Q5 ✓
CD3 - AT1 - Analytical Problem
Solving
20 marks

5/5 answered

```

2 int main()
3 {
4     int n,a[100],x,i,sort=1,c=0,found=0,l,r,m;
5     scanf("%d", &n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     scanf("%d", &x);
9
10    for(i=1;i<n;i++)
11        if(a[i]<a[i-1])
12            sort=0;
13    if(sort){
14        printf("Binary Search\n");
15        l=0;r=n-1;
16        while(l<=r){
17            m=(l+r)/2;
18            c++;
19            if(a[m]==x){
20                found=1;
21                break;
22            }
23            else
24                r=m-1;
25        }
26    }
27    else{
28        printf("Linear Search\n");
29        for(i=0;i<n;i++)
30            c++;
31        if(a[l]==x){
32            found=1;
33        }
34    }
35    if(found)
36        printf("Found\n");
37    else
38        printf("Not Found\n");
39    return 0;
40 }
41 }

```

5 2 8 1 9
8

Output

Linear Search
Not Found

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
C PROGRAMMING

 KAVIYAASRI M
192312614RB

Q3
CO3 - AT1 - Analytical Problem
Solving
20 marks

Q4
CO3 - AT1 - Analytical Problem
Solving
20 marks

Q5
CO3 - AT1 - Analytical Problem
Solving
20 marks

5/5 answered

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,m,a[50][50],i,j,s,b=0;
5     float t=0;
6     scanf("%d%d", &n,&m);
7     for(i=0;i<n;i++)
8         for(j=0;j<m;j++)
9             scanf("%d",&a[i][j]);
10    for(i=0;i<n;i++){
11        s=0;
12        for(j=0;j<m;j++)
13            s+=a[i][j];
14        if(s>b)
15            b=s;
16    }
17    printf("Topper Total: %d\n", b);
18    t=0;
19    for(i=0;i<n;i++)
20        for(j=0;j<m;j++)
21            t+=a[i][j];
22    printf("Class Average: %.2f\n", t/(n*m));
23    return 0;
24 }
```

Run Save

Input

```
3 3
80 70 90
60 85 75
95 88 92
```

Output

```
Topper Total: 275
Class Average: 81.67
```

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ENGINEERING

SIMATS Online Programming Lab
C PROGRAMMING

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Q3
CO3 - AT1 - Analytical Problem
Solving
20 marks

Q4
CO3 - AT1 - Analytical Problem
Solving
20 marks

Q5
CO3 - AT1 - Analytical Problem
Solving
20 marks

5/5 answered

d) Display strings beginning with vowels.

Your Code

```
1 #include<stdio.h>
2 #include<string.h>
3 int main() {
4     int n,i,j;
5     char a[20][50], t[50];
6     scanf("%d", &n);
7     for(i=0;i<n;i++)
8         scanf("%s", a[i]);
9     for(i=0;i<n;i++)
10        for(j=i+1;i<n;j++)
11            if(strcmp(a[i],a[j])>0{
12                strcpy(t,a[i]);
13                strcpy(a[i],a[j]);
14                strcpy(a[j],t);
15            }
16        for(i=0;i<n;i++)
17            printf("%s\n",a[i]);
18        return 0;
19    }
```

Run Save

Input

```
5
banana
apple
orange
```

Output

Q2
CO3 - AT1 - Analytical Problem
Solving
20 marks

Q3
CO3 - AT1 - Analytical Problem
Solving
20 marks

Q4
CO3 - AT1 - Analytical Problem
Solving
20 marks

Q5
CO3 - AT1 - Analytical Problem
Solving
20 marks

5/5 answered

Example:

Input:

aaabbbcccccdd

Output:

a3b3c4d2

Also calculate the compression ratio.

Your Code

```
1 #include<stdio.h>
2 int main() {
3     char s[100];
4     int i=0, c;
5     scanf("%s", s);
6     while(s[i]!='\0') {
7         c=1;
8         while(s[i]==s[i+1]) {
9             c++;
10            i++;
11        }
12        printf("%c%d", s[i],c);
13        i++;
14    }
15    return 0;
16 }
```



Run



Save

Input

aaabbbcccccdd

Output

scanf("%s",...) without a width
limit is not allowed - use
scanf("%99s",...) or fgets()
instead.